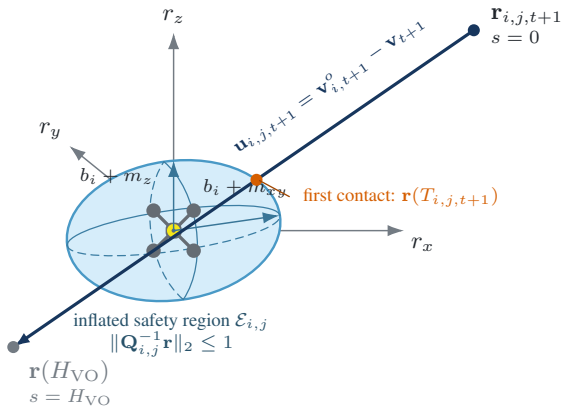
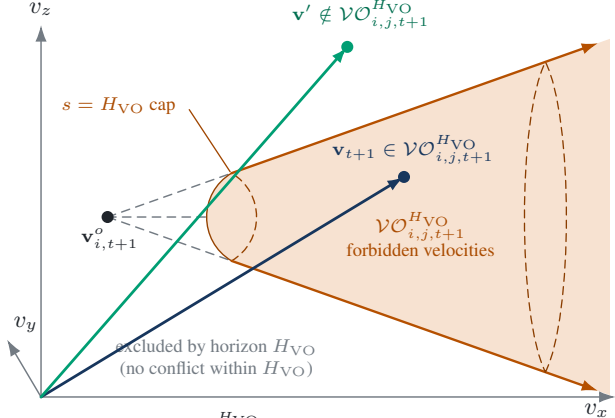




$$\mathbf{r}(s) = \mathbf{r}_{i,j,t+1} + s\mathbf{u}_{i,j,t+1},$$

$$T_{i,j,t+1} = \min\{s \in (0, H_{VO}] : \mathbf{r}(s) \in \mathcal{E}_{i,j}\}$$

(a) Relative-space TTC test



$$\mathbf{v} \in \mathcal{VO}_{i,j,t+1}^{H_{VO}} \iff \exists s \in (0, H_{VO}] :$$

$$\|\mathbf{Q}_{i,j}^{-1}[\mathbf{r}_{i,j,t+1} + s(\mathbf{v}_{i,t+1}^o - \mathbf{v})]\|_2 \leq 1$$

(b) Finite-horizon 3D velocity obstacle